

OSTN15 NTV2 TRANSFORMATION

Data Format and User Guide

OSTN15 NTv2 Transformation

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Contents

Section

Preface **3**
Contact details 3
Use of the transformation model 3
Disclaimer 3
Copyright in this guide 4
Data copyright and other intellectual property rights 4
Trademarks 4
Using this guide 4
Chapter 1 **Introduction** **5**
OSTN15 NTv2 format 5
ETRS89 explained 5
Benefits 6
Applications 6
Software for OSTN15 6
Chapter 2 **NTv2 Data Format** **7**
Basic principles 7
Data structure 7
Chapter 3 **Using OSTN15 NTv2** **9**
Chapter 4 **Quality statement** **10**
Coverage 10
Accuracy of OSTN15 10
Annexe A **Glossary** **11**

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Preface

This guide is designed to provide an overview of the NTv2 format version of the Ordnance Survey grid transformation in Great Britain OSTN15™. It is a guide to help users understand the format of the transformation data file only. Since the implementation of the NTv2 will most likely involve the binary version of the data file in a GIS system the interpolation method and formulae are not covered. This guide assumes that users have an understanding of coordinate systems and datums. If you find an error or omission in this user guide, or otherwise wish to make a comment or suggestion as to how we can improve the user guide, please contact us at the address shown below under contact details.

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Use of the transformation model

Within Great Britain coordinates are transformed using the Ordnance Survey National Grid Transformation model (OSTN15).

The OSTN15 transformation model (in “native” and NTv2 formats) has been created by Ordnance Survey Great Britain who are the organisation responsible for the official, definitive topographic mapping of the country.

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Using this guide

The documentation is supplied in portable document format (PDF) only. Free Adobe® Reader® software, which displays the specification, incorporates search and zoom facilities and allows you to navigate within. Hyperlinks are used to navigate between associated parts of the specification and to relevant internet resources by clicking on the blue hyperlinks and the table of contents.

Chapter 1 Introduction

All Ordnance Survey mapping relates to a coordinate reference system known as OSGB36® (the "National Grid" – EPSG code 27700). This reference system was traditionally realised on the Earth's surface by monumented triangulation stations. The users of mapping products, in both the public and private sectors, have invested in geographical information systems (GIS) and asset management systems based on OSGB36 which has been accepted as de facto national standard.

The National Grid is capable of supporting surveying and mapping in GB to meet all the requirements of users both now and in the future; however, an increasing number of spatial datasets are available in GNSS (Global Navigation Satellite System, e.g. GPS) compatible coordinate systems, such as ETRS89 (EPSG code 4258). When two or more coordinate datasets are to be integrated, it is essential that each relates to the same coordinate reference system, irrespective of accuracy issues.

In order to relate GNSS-derived positions to Ordnance Survey's mapping, GNSS coordinates need to be transformed to National Grid via a specialised datum transformation. For this reason Ordnance Survey has developed OSTN15, the standard datum transformation for Great Britain. OSTN15 replaces the previous model OSTN02™.

In its "native" form OSTN15 consists of a 700km by 1,250km square grid of translation vectors at 1km resolution. This provides a fit between the GNSS coordinate system European Terrestrial Reference System 1989 (ETRS89) and the OSGB36 National Grid. OSTN15 is in agreement with major triangulation stations at the level of 0.1m root mean square error (RMSE).

OSTN15 has been developed from the national primary, secondary and tertiary triangulation station network. It contains over 3,200 points directly observed by GNSS and more than 1,000 from the original retriangulation observations adjusted on the ETRS89 datum.

OSTN15, in conjunction with the ETRS89 positions of the OS Net® permanent GNSS stations, is now the official definition of OSGB36 National Grid coordinate system. This means that using OSTN15 with the OS Net Network, surveyors using GNSS have no need to occupy triangulation stations in order to relate GNSS coordinates to National Grid coordinates.

OSTN15 NTv2 format

The format in which OSTN15 is originally derived, from the raw data set of observed triangulation points, is a regular 1km square grid of parameters, 700km wide and 1250km high covering the whole of GB. Transformation parameters at a particular point are computed via bi-linear interpolation from the parameters at the corners of the km square within which the point falls. The file format of OSTN15 is a simple comma delimited text file with each line containing the parameters of a single km square intersection along the coordinates of that intersection.

NTv2 transformation format is popular with GIS systems and follows a similar principle using a grid of parameters and bi-linear interpolation. The main difference from native OSTN15 format is that NTv2 is a latitude/longitude graticule instead of a km square grid. The file format for OSTN15 NTv2 is more prescribed than for native OSTN15. There is a text file (extension *.asc*) with a header block that gives basic information about the transformation and then the parameters are given in a fixed character width, space delimited, column format with one line for each graticule intersection giving just the shifts and their estimated error values. The other difference from native OSTN15 is that the NTv2 version is offered in two versions – one with native direction ETRS89 to OSGB36 (the same as native OSTN15) and another where the native direction is reversed to be OSGB36 to ETRS89. The transformation (in both formats) is fully reversible. NTv2 also has a binary companion to the *.asc* file with the extension *.gsb*. It is the *gsb* file that is most commonly used in GIS systems.

ETRS89 explained

OSTN15 links the OSGB36 National Grid coordinate reference system to the GNSS-compatible coordinate system ETRS89. In Europe, ETRS89 is a precise version of the better known WGS84 (EPSG code 4326) coordinate reference system optimised for use in Europe; however, for most purposes it can be considered equivalent to WGS84.

Specifically, the motion of the European continental plate is not apparent in ETRS89, which allows a fixed relationship to be established between this system and Ordnance Survey mapping coordinate systems.

Additional precise versions of WGS84 are also currently in use, notably ITRS (International Terrestrial Reference System); these are not equivalent to ETRS89. The difference between ITRS/WGS84 and ETRS89 was zero in 1989 and is growing by approximately 0.025 m per year in UK and Ireland (so is ~0.67m in 2016). This effect is only relevant in international scientific applications. For all navigation, mapping, GIS, and engineering applications within the tectonically stable parts of Europe (including UK and Ireland), the term ETRS89 should be taken as synonymous with WGS84.

Benefits

OSTN15 provides the solution to relating GNSS (WGS84) datasets to Ordnance Survey mapping. Used with the OS Net GNSS network, it allows GNSS surveying within the National Grid without the need to visit any Ordnance Survey traditional control points.

Applications

The OSTN15 is of interest to:

- GNSS surveyors who need to relate their survey to the National Grid – used with OS Net, it removes the need to visit traditional Ordnance Survey horizontal and vertical control points; and
- GIS, GPS, CAD and navigation system developers who need to integrate GNSS (WGS84) datasets with Ordnance Survey mapping – these products provide the complete solution to these users at all Ordnance Survey mapping scales.

Software for OSTN15

This guide specifically deals with the OSTN15 NTv2 data format which is designed for incorporation into GIS Systems. However, the transformation (and its companion height model OSGM15) have been coded into a software application – "Grid InQuest II". The software allows for individual coordinate input and output via a GUI and also batch input/output via text files. A command line interface and dll, along with examples of their use in a variety of programming languages, are also included. Users wishing to incorporate the pre-prepared .dll into other applications should refer to the Grid InQuest II user guide.

Grid InQuest II download packages for Windows® (32 bit and 64 bit), Linux® (32 bit and 64 bit) and OSX® are available from the Ordnance Survey web *site http://www.ordnancesurvey.co.uk/business-and-government/help-and-support/navigation-technology/os-net/grid-inquest.html*. The packages and software source code are also available at *https://bitbucket.org/PaulFMichell/gridinquestii*

Chapter 2 NTv2 Data Format

Basic principles

Specifications of OSTN15 horizontal transformation, NTv2 format

Transformation:	horizontal datum transformation between OSGB36 and ETRS89. **Native direction of parameters is indicated by the data file name**, i.e. "OSTN15_NTv2_OSGBtoETRS" or "OSTN15_NTv2_ETRStoOSGB" but transformation is reversible.
Transformation type:	Interpolation from graticule of latitude and longitude shifts (in seconds).
Estimation method:	Bi-linear interpolation from native OSTN15 1km grid.
Graticule resolution:	30" in latitude, 60" in longitude.
Grid interpolation:	Bi-linear.
Accuracy:	0.1m (RMS) with respect to OSGB36 primary, secondary and tertiary triangulation monuments and 0.001m with respect to native 1km grid version of OSTN15.
Extent:	SW corner N49°, W9°; NE corner N61°, E2°. 12° North-South, 11° East-West.

Data structure

The data structure of the *.asc* file of OSTN15 NTv2 is shown below, **however most practical applications of OSTN15 NTv2 will involve importing the binary *.gsb* file**. Usually when using the *.gsb* file no knowledge of the data structure is required and it is normally sufficient to just state the native 'from' and 'to' coordinate systems of the transformation.
For OSTN15 NTv2 the native 'from' system and the 'to' system is indicated by the *.asc* or *.gsb* filename. I.e. *OSTN15_NTv2_OSGBtoETRS.asc* or *OSTN15_NTv2_ETRStoOSGB.asc*. The example data structure shown here is of native direction OSGB36 to ETRS89.

The data file starts with a header block:

Header element	Description	Value		
NUM_OREC 11	No. records in overview header	11	1.	Start of overview header
NUM_SREC 11	No. records in sub grid(s) header	11	2.	
NUM_FILE 1	No. sub grids	1	3.	
GS_TYPE SECONDS	Shift type	SECONDS	4.	
VERSION OSTN15	Model name	OSTN15	5.	
SYSTEM_FAIRY1830	"From" ellipsoid	AIRY1830	6.	
SYSTEM_TGRS80	"To" ellipsoid	GRS80	7.	
MAJOR_F 6377563.396	"From" ellipsoid semi major axis (m)	6377563.396	8.	
MINOR_F 6356256.909	"From" ellipsoid semi minor axis (m)	6356256.909	9.	
MAJOR_T 6378137.000	"To" ellipsoid semi major axis (m)	6378137.000	10.	
MINOR_T 6356752.314	"To" ellipsoid semi minor axis (m)	6356752.314	11.	End of overview header
SUB_NAMEGB+IOM	Sub grid name	GB+IOM	1.	Start of sub grid header
PARENT NONE	Parent grid name	NONE	2.	
CREATED 20161003	Creation date (YYYYMMDD)	20161003	3.	
UPDATED 20161003	Updated date (YYYYMMDD)	20161003	4.	
S_LAT 176400.000000	Lower latitude (seconds)	176400 (N49°)	5.	
N_LAT 219600.000000	Upper latitude (seconds)	219600 (N61°)	6.	
E_LONG -7200.000000	Lower longitude (seconds, W +ve)	-7200 (E2°)	7.	
W_LONG 32400.000000	Upper longitude (seconds, W +ve)	32400 (W9°)	8.	
LAT_INC 30.000000	Latitude interval (seconds)	30	9.	
LONG_INC 60.000000	Longitude interval (seconds)	60	10.	
GS_COUNT952501	No. grid nodes	952501	11.	End of sub grid header

The header is then immediately followed by the rows of data. Each row contains the offsets for a single graticule intersection or "node". The node order in the file is indicated in Figure 1 below.

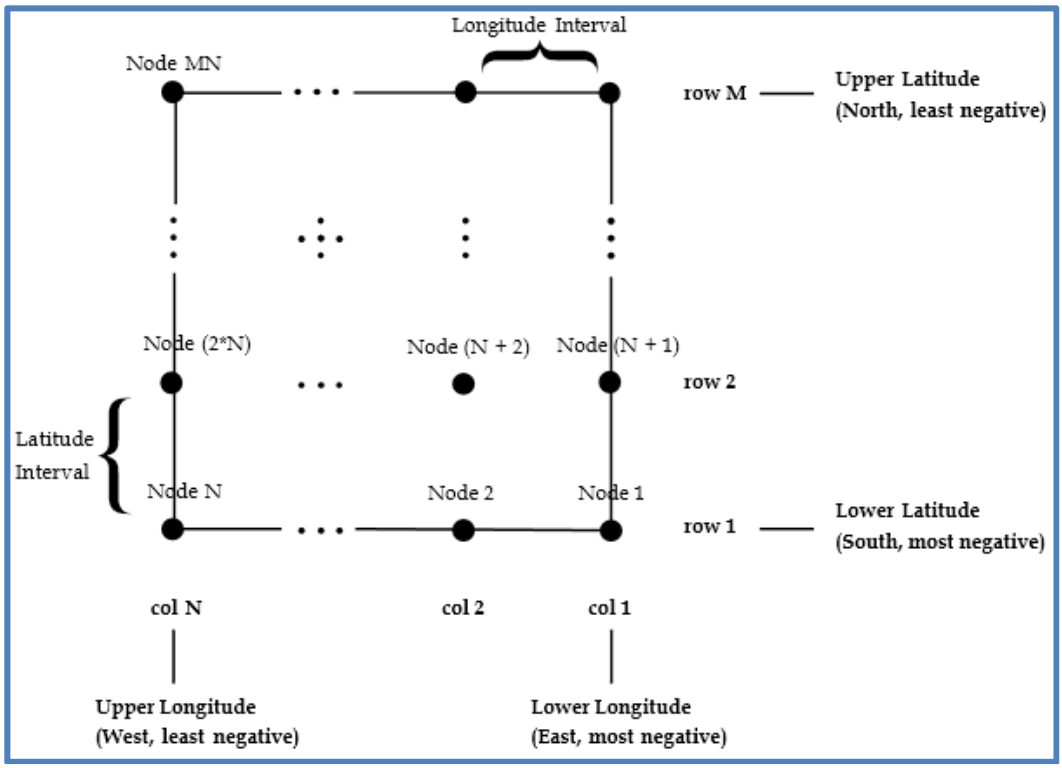


Figure 1. Diagram of NTv2 file grid node order (from https://www.dnrm.qld.gov.au/__data/assets/pdf_file/0006/105765/gday-21-user-guide.pdf).

So, the order for OSTN15 NTv2 format is as follows:

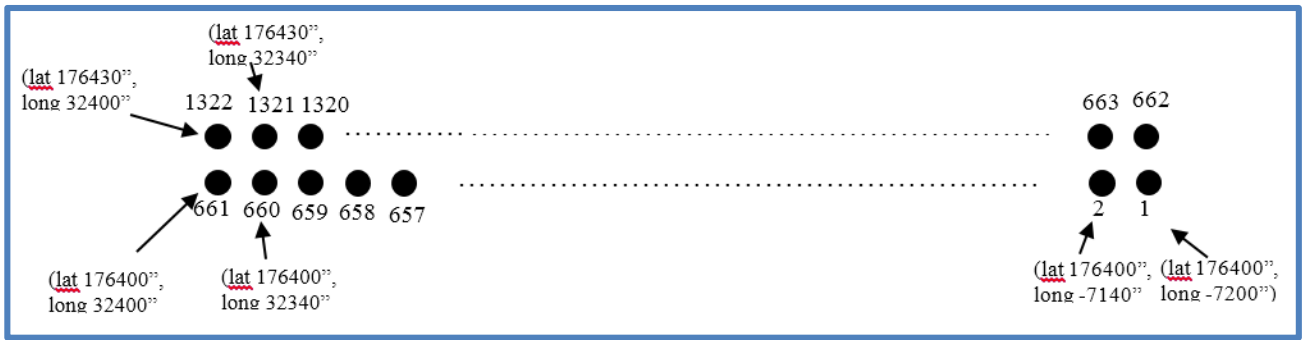


Figure 2. Nodes of the OSTN15 NTv2 data.

The data rows look like this:

```
  2.998631   6.259749   0.000030   0.000030
  2.997586   6.253345   0.000030   0.000030
  2.996542   6.246941   0.000030   0.000030
  2.995498   6.240536   0.000030   0.000030
.
.
.
```

Each row is a latitude shift value (seconds) and longitude shift value (seconds) followed by accuracy indicators in the same order (latitude then longitude).

The binary *.gsb* files for OSTN15 NTv2 were generated from the *.asc* files using the "GDAy 2.1" software available from this web page:
https://www.business.qld.gov.au/industry/titles-property-construction/surveying/gda-transformation-software

Chapter 3 Using OSTN15 NTv2

The use of OSTN15 NTv2 will most likely involve introducing the binary *.gsb* file into a GIS system. So, the formula and algorithms for interpreting the NTv2 *.asc* file are not give here. Users interested in the formula and algorithms for bilinear interpolation of transformation parameters from a gridded file should see the user guide for native OSTN15 - "Transformations_and_OSGM15_User_Guide" which is available in the *OSTN15-OSGM15_Developers_Pack.zip* from the Ordnance Survey web site.

Specific instructions for installing the OSTN15 NTv2 transformation in different GIS systems are beyond the scope of this document and users are advised to contact their GIS system vendor for details. However the process of using OSTN15 NTv2 *.gsb* file in a GIS will usually involve specifying the "from" and "to" coordinate systems. **For the OSTN15 NTv2 *.gsb* files the 'from' system and the 'to' system are indicated by the filename. I.e. *OSTN15_NTv2_OSGBtoETRS.gsb* or *OSTN15_NTv2_ETRStoOSGB.gsb*.**

The coordinate system and ellipsoid information that may be required by some GIS systems are given in Table 1 below.

Table 1. Coordinate system information for OSTN15 NTv2

Coordinate system, EPSG code	**Ellipsoid**	**Semi-major axis *a* (m)**	**Semi-minor axis *b* (m)**
OSGB36 (2D lat/long), EPSG::4277 or OSGB36 National Grid, EPSG::27700	Airy 1830	6 377 563.396	6 356 256.909
ETRS89, EPSG::4258 (or WGS84, EPSG::4326)	GRS80[1]	6 378 137.000	6 356 752.3141

The *OSTN15_NTv2.zip* package available from the Ordnance Survey web site contains the the *.asc* and *.gsb* data files along with test input and test output files to allow for testing of the implementation of OSTN15 NTv2 in GIS systems. Sample test inputs and outputs are given in Table 2 and Table 3 below. NB the exact answers below may not be produced due to differences between GIS systems in data handling and interpretation of the transformation algorithm. However results should be with a cm or so.

Table 2. Example OSGB36 to ETRS89 transformation

OSGB36 National Grid, EPSG::27700 INPUT (m)	ETRS89, EPSG::4258 OUTPUT (Decimal Degrees)
`422242.186, 433818.701`	`53.80021519630,-1.66379168242`

Table 3. Example ETRS89 to OSGB36 transformation.

ETRS89, EPSG::4258 INPUT (Decimal Degrees)	OSGB36 National Grid, EPSG::27700 OUTPUT (m)
`51.40078220140, -3.55128349240`	`292184.870, 168003.465`

[1] Also known as the WGS84 ellipsoid

Chapter 4 Quality statement

Coverage

OSTN15 covers Great Britain and the Isle of Man.

Accuracy of OSTN15

Within Great Britain, OSTN15 is the definitive OSGB36/ETRS89 transformation. OSTN15 in combination with the ETRS89 coordinates of the OS Net GNSS network stations, rather than the fixed triangulation network, now define the National Grid. This means that, for example, the National Grid coordinates of an existing OSGB36 point, refixed using GNSS from OS Net and OSTN15, will be the correct ones. The original archived OSGB36 National Grid coordinates of the point (if different) will no longer be true OSGB36, by definition, but the two coordinates (new and archived) will agree on average to better than 0.1 m, (68% probability).

In NTv2 format the above accuracy statement also applies with the further statement that the NTv2 version of OSTN15 agrees with the native 1km grid version of OSTN15 at the 0.001m level (i.e. the difference between native OSTN15 and OSTN15 NTv2 is negligible).

Annexe A Glossary

The following is a list of technical terms that maybe found in this user guide, together with a fuller definition.

datum

A point, line, surface or set of these, with respect to which positions of objects can be stated as unique sets of coordinates.

de facto national standard

A national standard by adoption rather than legally enforced.

ellipsoid (biaxial)

The 3D geometric figure obtained by rotating an ellipse about its minor axis. Used in geodesy to approximate the shape of the earth.

ETRF89

European Terrestrial Reference Frame 1989 – the Europe-fixed realisation of WGS84. Governed by EUREF as a standard reference frame for Europe.

ETRS89

European Terrestrial Reference System 1989 – a coordinate system that is the Europe-fixed precise version of WGS84. Governed by EUREF as the standard fixed reference system for Europe. ETRS89 is related to the state-of-the-art WGS84-consistent system ITRS2000 by a six-parameter kinematic transformation published by IERS.

EUREF

EUREF (European Reference Frame): a sub-commission of the International Association of Geodesy, Commission X.

geocentric datum

A reference system that uses the centre of mass of the earth as its origin; the popularity of these systems today derives from their usefulness in describing satellite orbits.

Geoid model

A model of the level surface which is closest to mean sea level over the oceans. This surface is continued under land as the fundamental reference surface for height measurement.

GNSS

Global Navigation Satellite System – generic term for one or more satellite navigation systems including (but not limited to) GPS (USA), GLONASS (Russia), Galileo (Europe), Beidou (China).

GPS

Global Positioning System – an outdoor positioning technique using a constellation of US Department of Defense satellites and a portable receiver to dynamically determine coordinates. For high precision, several receivers are used and their relative positions are determined.

GRS80

A global reference ellipsoid used in the WGS84 coordinate system. Also known as the WGS84 ellipsoid.

IERS

International Earth Rotation Service.

ITRF

International Terrestrial Reference Frame – the state-of-the-art global realisation of the WGS84 reference system, using observations from worldwide networks of active geodetic stations of the VLBI, SLR, GPS and DORIS techniques.

ODN

Ordnance Datum Newlyn – the levelling-based vertical reference frame for most of the British Isles, with a single tide gauge constraint in Newlyn in Cornwall.

OSGB36

Ordnance Survey Great Britain 1936 – the British horizontal mapping datum, observed by triangulation from 1936 and traditionally realised on the ground by triangulation stations. With the release of the definitive transformation, OSTN02, OSGB36 is now realised by the ETRS89 coordinates of the National GPS Network in conjunction with the OSTN02 transformation.

OSGM02

Ordnance Survey National Geoid Model 2002 – now superceded, a gravimetric Geoid model that is aligned with the national height datums of Great Britain, Northern Ireland and Ireland. Replaced by OSGM15.

OSGM15

Ordnance Survey National Geoid Model 2015 – a gravimetric Geoid model that is aligned with the national height datums of Great Britain, Northern Ireland and Ireland.

OSTN15

Ordnance Survey National Grid Transformation 2015 – a grid shift type horizontal transformation between the ETRS89 datum and OSGB36 National Grid.

realisation

A spatial reference system made real on the ground by monumented points with estimated coordinates and errors.

transformation

A procedure to change from one coordinate system to another.

WGS84

World Geodetic System 1984 – the global geodetic reference system used to describe the position of GPS satellites and ground stations.